

BIOS vs UEFI: A Technical Comparison

Week 3 Portfolio Project, Part C

ITEP 414, System Administration and Maintenance

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1. Introduction

The firmware of a computer is the first software that runs when the machine is powered on. Its job is to initialize the hardware, run self-tests, and hand control over to the operating system. For decades, the Basic Input/Output System, or BIOS, performed this task. In recent years, however, the Unified Extensible Firmware Interface, or UEFI, has taken its place as the standard firmware on all new computers. This document compares the two technologies and explains why UEFI has largely replaced BIOS in modern systems.

2. Comparison Table

Aspect	BIOS	UEFI
Definition	Legacy firmware stored on a ROM chip on the motherboard; initializes hardware and loads the boot loader from the Master Boot Record (MBR).	Modern firmware interface that performs the same role but is written in modular C code, stored on flash memory, and loads boot loaders as .efi applications.
Boot Process	Runs POST (Power-On Self-Test), then reads the first 512 bytes of the boot disk (the MBR) and executes the boot loader found there.	Runs initialization in parallel, then uses the EFI boot manager to read boot loader files (for example, grubx64.efi) from the EFI System Partition.
Maximum Disk Support	2 TB. Disks larger than this cannot be fully addressed because the MBR partition table uses 32-bit entries.	Effectively unlimited (the GPT partition table supports up to 9.4 zettabytes), so multi-terabyte drives work without workarounds.
Partition Style	MBR (Master Boot Record), which supports a maximum of four primary partitions.	GPT (GUID Partition Table), which supports up to 128 partitions per disk and stores redundant copies of the table for safety.
Security Features	None built in. Any boot loader can be loaded, which allows unsigned or malicious code to take over the boot process.	Secure Boot verifies the digital signature of the boot loader before running it, blocking unsigned and untrusted code.
Speed	Slower. Hardware is initialized one device at a time, which makes the boot process longer.	Faster. Devices are initialized in parallel, and the interface can also support faster resume from standby.
Advantages	Simple design, works with very old operating systems, and easy to understand and troubleshoot.	Fast boot, support for very large disks, Secure Boot, a graphical setup interface, driver support, and firmware updates

		without replacing a chip.
Disadvantages	2 TB disk limit, no security features, slow initialization, and a text-only interface.	More complex, requires GPT disks, and older operating systems may not boot if Secure Boot is enabled.
Modern Usage	Kept mainly for compatibility with old operating systems and legacy hardware; most systems emulate it inside UEFI.	Standard on every desktop, laptop, and server manufactured since about 2012; required for current versions of Windows.

Table 1. Side-by-side comparison of BIOS and UEFI firmware.

3. Why UEFI Has Replaced BIOS

UEFI has replaced BIOS for practical reasons that matter to both manufacturers and users. The first reason is disk capacity. Storage drives have grown well past the 2 TB limit that BIOS and MBR impose, and the modern GPT partition table used by UEFI removes that barrier entirely. A server administrator installing an 8 TB data drive simply cannot do it cleanly with a BIOS system, so choosing UEFI is almost mandatory in enterprise environments.

The second reason is security. BIOS runs whatever boot loader it finds, with no verification. That makes a computer vulnerable to bootkits, which are malware that hide inside the boot sector and survive reinstallation of the operating system. UEFI addresses this with Secure Boot, a feature that checks the digital signature of every boot loader before execution and refuses to run anything unsigned. For organizations that handle sensitive data, this protection is no longer optional.

The third reason is speed and usability. UEFI initializes hardware in parallel rather than one device at a time, which shortens the time from power-on to login. It also provides a graphical setup utility with mouse support, built-in diagnostics, and network boot tools that BIOS cannot offer. Firmware updates are delivered as ordinary files and applied through the interface, instead of requiring a chip replacement.

Finally, the market itself pushed the change. In 2012, Microsoft required UEFI support, including Secure Boot, for hardware that wanted Windows 8 certification, and every major vendor followed. Linux distributions now ship with UEFI support as the default. Because every new motherboard, laptop, and server ships with UEFI firmware, BIOS survives only as a compatibility mode inside UEFI, and new hardware no longer includes it at all. For these reasons, UEFI is now the standard boot technology in modern computers, and BIOS is effectively retired except for legacy use.